RAINFALL FORECASTING

Exploratory Data Analysis and Comparative Deep-Learning Study

Course Code: CSE638

Course Name: Deep Learning

Topic Name: A study of next-day rainfall prediction using BiLSTM, BiLSTM-PINN, and Transformer model*s* (Task-4)

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Abstract

Rainfall forecasting is challenging because daily precipitation contains many zero-rain observations, a strongly seasonal pattern, substantial differences between locations, and a small number of high-impact extreme events. This project develops and evaluates a station-level, next-day rainfall forecasting pipeline using the Indian Rainfall and Weather Prediction Dataset collected through the Meteostat API. The supplied dataset description reports more than 970,000 records from 458 weather stations over 1 January 2015–10 February 2025. To align the exploratory analysis with the modelling pipeline and reduce bias caused by non-random missingness in early years, the experiments retain complete records from 2 January 2021 onward. The resulting cleaned data contain 597,294 station-day rows and 414 station identifiers.

Each model receives a 30-day sequence of 21 weather, geographic, seasonal, and regional features and predicts rainfall for the following day. The study compares a bidirectional Long Short-Term Memory network (BiLSTM), a BiLSTM with physics-informed auxiliary constraints (BiLSTM-PINN), a Transformer encoder, a histogram gradient-boosting reference model, and two baselines. Neural networks are trained on log1p-transformed rainfall; predictions are converted back to millimetres and calibrated using validation data. On 101,540 held-out test windows from 412 stations, calibrated BiLSTM-PINN achieved the lowest neural-model RMSE (10.491 mm; $R^2 = 0.369$), narrowly ahead of calibrated BiLSTM and Transformer. Differences between the three neural models are small, and the results show that extreme-rainfall prediction remains difficult.

Keywords: rainfall forecasting, time-series modelling, BiLSTM, physics-informed learning, Transformer, Meteostat, weather prediction.

Report Roadmap

Section	Content
1–2	Introduction, problem statement, objectives, and research questions
3–4	Dataset provenance, data audit, cleaning, and chronological partitioning
5–6	Feature engineering, sequence generation, and exploratory analysis
7	Model architectures and physics-informed loss formulation
8–9	Training procedure, baselines, evaluation metrics, and results
10–12	Error analysis, limitations, conclusion, and references

1. Introduction

Rainfall information is important for crop planning, irrigation, reservoir operation, flood preparedness, and broader climate analysis. A forecast that estimates rainfall at a station for the next day can support local decisions, provided that its limitations are understood. However, rainfall is not a simple continuous signal: many station-days are dry, rainfall amounts are non-negative, the distribution has a long upper tail, and weather conditions vary geographically and seasonally.

Classical persistence and climatology approaches offer useful reference points, but they do not fully represent temporal dependencies across multiple days or interactions between nearby stations. Recurrent neural networks can model sequential information, while Transformer encoders use attention to represent

relationships across a historical window. Physics-informed constraints can additionally discourage outputs that contradict basic physical expectations.

The present work is an empirical comparison rather than a claim that any one architecture is universally optimal. We apply the same data preparation, input windows, chronological test period, and core evaluation protocol to three neural approaches, and compare them with simpler baselines and a gradient-boosting reference. The goal is to determine how accurately these models forecast next-day rainfall in millimetres and how well they identify rainy days.

1.1 Problem Statement

Given the recent weather history of a station and context from its geographical surroundings, estimate the rainfall amount at that station on the next calendar day. The model must be evaluated on later dates than those used for fitting, without randomly mixing adjacent observations across the training and test sets.

$$\mathbf{X}_t = [\mathbf{x}_{t-29}, \mathbf{x}_{t-28}, \ldots, \mathbf{x}_t] \rightarrow \hat{\mathbf{y}}_{t+1} \approx \mathbf{rainfall}_{t+1}\ \mathbf{(mm)}$$

A 30-day input sequence is used to predict the rainfall for the next day.

1.2 Objectives

- Audit and clean the daily station data using explicit, reproducible rules.
- Explore missingness, rainfall distribution, seasonality, spatial variability, and relationships between features and next-day rainfall.
- Construct leakage-aware features and valid 30-day sequences for one-day-ahead prediction.
- Train and compare BiLSTM, BiLSTM-PINN, and Transformer models under a common chronological evaluation setup.
- Benchmark the neural models against climatology, persistence, and gradient boosting.
- Evaluate both rainfall amounts and rain/no-rain discrimination, and analyse performance by rainfall intensity, month, and station.

1.3 Research Questions

- RQ1: Do sequence-based neural models improve next-day rainfall estimates compared with climatology and persistence?
- RQ2: Does adding physics-informed auxiliary constraints improve rainfall prediction relative to a standard BiLSTM?
- RQ3: How similar are the BiLSTM, BiLSTM-PINN, and Transformer results on the same held-out test period?
- RQ4: How does performance vary with rainfall intensity, month, and station location?

2. Dataset and Data Provenance

The project uses the “Indian Rainfall and Weather Prediction Dataset,” described by its provider as a structured time-series dataset collected using the Meteostat API and Python library. The dataset description reports daily observations for 458 weather stations across India from 1 January 2015 to 10 February 2025, with more than 970,000 records. The source documentation is available at https://dev.meteostat.net/.

Category	Fields / meaning
Weather measurements	Average, minimum, and maximum temperature; average wind speed; average air pressure; total daily rainfall/precipitation.
Location	Station name, state, district, latitude, longitude, and elevation.
Time	Date of record, month, year, and season; day-of-year is also transformed into cyclical variables for modelling.
Target variable	Daily rainfall in millimetres. The prediction target is rainfall on the day after the 30-day input window.

The model reads the Excel file `india_weather_rainfall_data.xlsx`. Its raw data audit reports approximately 970 thousand rows and a record period extending to 10 February 2025. Then creates a station key using station name, latitude, longitude, and elevation. This is necessary because station names alone are not unique.

Important distinction: the dataset description reports 458 stations in the original resource, while the cleaned analysis and evaluation use the station identifiers that remain after applying the project's quality rules. The final cleaned table contains 414 station identifiers; the held-out test set contains predictions for 412 stations.

2.1 Data Licence

The supplied dataset description specifies the Creative Commons Attribution–NonCommercial 4.0 International licence (CC BY-NC 4.0). Reuse should credit the dataset/Meteostat source and comply with the non-commercial restriction. The project should not imply that the dataset is unrestricted for commercial use.

2.2 Scope of the Experiments

Although the original resource begins in 2015, the modelling experiments do not use the full period. They start on 2 January 2021 because the audit identifies extensive, seasonally non-random missingness before 2021. All subsequent statistics and model scores in this report refer to the cleaned records and the sequence windows actually used, not to every row in the original download.

3. Data Audit and Preprocessing

3.1 Audit of Missing Values

The first step checks the number and proportion of missing values in rainfall, wind speed, air pressure, maximum temperature, and minimum temperature by year. The audit shows that missingness is particularly severe during 2015–2020. For example, the proportion of rows with missing rainfall is approximately 67–71% in 2015–2020, compared with about 0–3% in 2021–2025. Wind and pressure are also frequently missing in the early period.

This missingness is not random: missing rainfall in the early years is concentrated in dry months. If incomplete rows were simply deleted across the entire 2015–2025 period, the retained early-period data would over-represent rainy days and distort the rainfall distribution. Therefore, the project begins modelling from 2 January 2021, when completeness is substantially better.

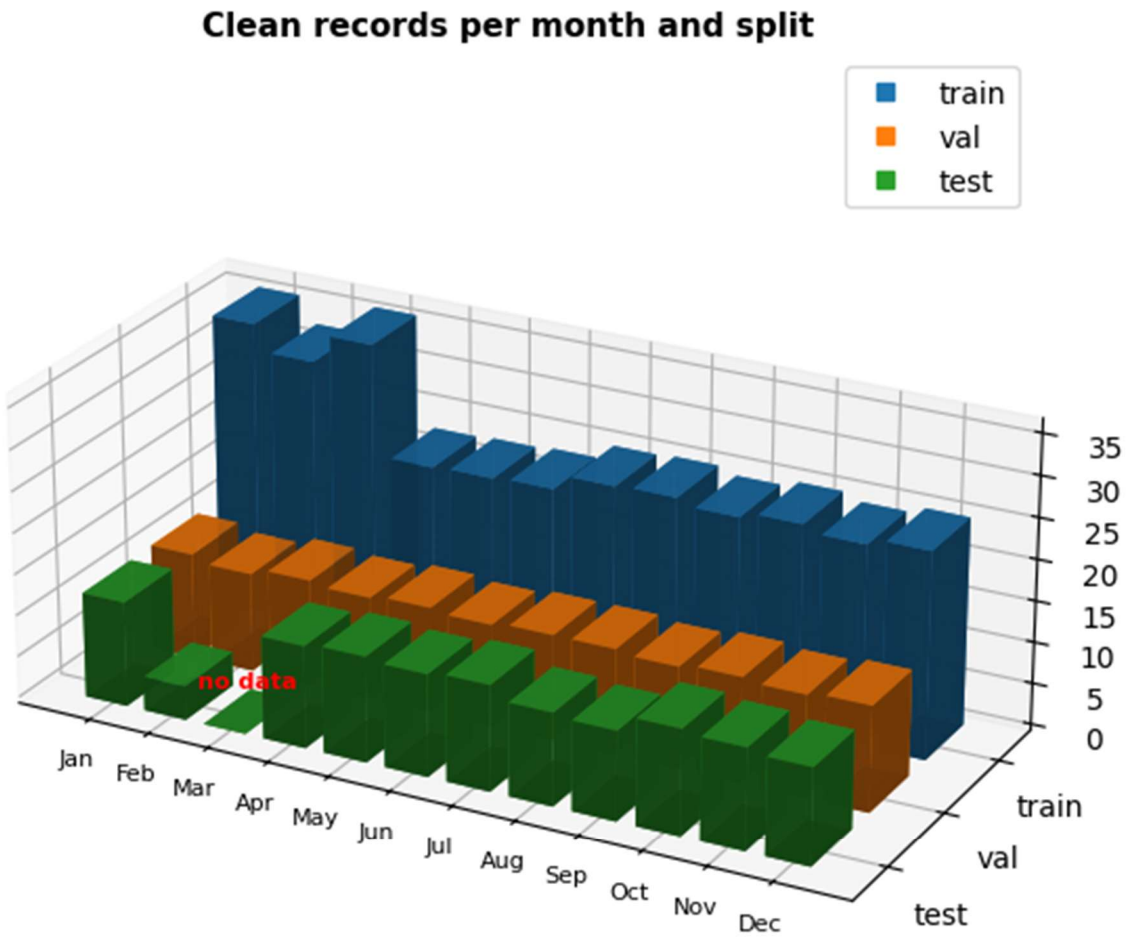


Figure 1. Data audit and cleaning summary

Interpretation: The audit highlights the amount and pattern of missing observations before modelling. Missingness is not random across the full historical period, so the pipeline uses explicit cleaning rules and a restricted modelling period instead of treating every row as equally reliable.

3.2 Cleaning Rules

Step	Rule applied	Reason
Date filter	Keep records on or after 2021-01-02.	Avoid the early period with extensive, non-random missingness.
Missing / non-finite values	Delete any row with a missing required field or a non-finite model variable. No general imputation is applied to the core weather rows.	Prevent missing values from entering model windows.
Temperature validity	Delete rows with maximum temperature greater than 50 °C.	Remove values treated as physically implausible.
Station identifier	Combine station name, rounded latitude, longitude, and elevation.	Avoid merging different physical stations with the same name.

Rainfall handling	Do not convert missing rainfall into 0 mm; use log1p after clipping negative rainfall values to zero in feature construction.	A missing observation is not equivalent to a measured dry day.

After applying the cleaning rules, it retains 597,294 station-day records across 414 station identifiers in 32 states/union territories. The row counts by chronological split are shown below.

Split	Date range	Clean records	Stations represented	Share of clean rows
Training	2021-01-02 – 2023-03-31	321,063	412	53.8%
Validation	2023-04-01 – 2024-03-31	150,294	413	25.2%
Test	2024-04-01 – 2025-02-10	125,937	412	21.1%
Total	2021-01-02 – 2025-02-10	597,294	414 unique station IDs	100%

3.3 Chronological Data Split

The split is defined by calendar dates rather than random sampling. Training covers 2 January 2021–31 March 2023, validation covers the complete year 1 April 2023–31 March 2024, and the final test covers 1 April 2024–10 February 2025. A full year is reserved for validation so that it includes all months. The test period ends in February and contains no March observations; therefore, test metrics describe the dates available in that interval and should not be interpreted as a balanced comparison of every calendar month.

The model sets `CONTEXT_FROM_PREVIOUS_SPLIT = True`. This means that each target must belong to its designated split, but the historical input window is allowed to use the immediately preceding period when forecasting at a split boundary. This matches a realistic forecasting situation: previously observed days are available when a forecast is made. The targets remain chronologically separated, and no future observations are included in an input sequence.

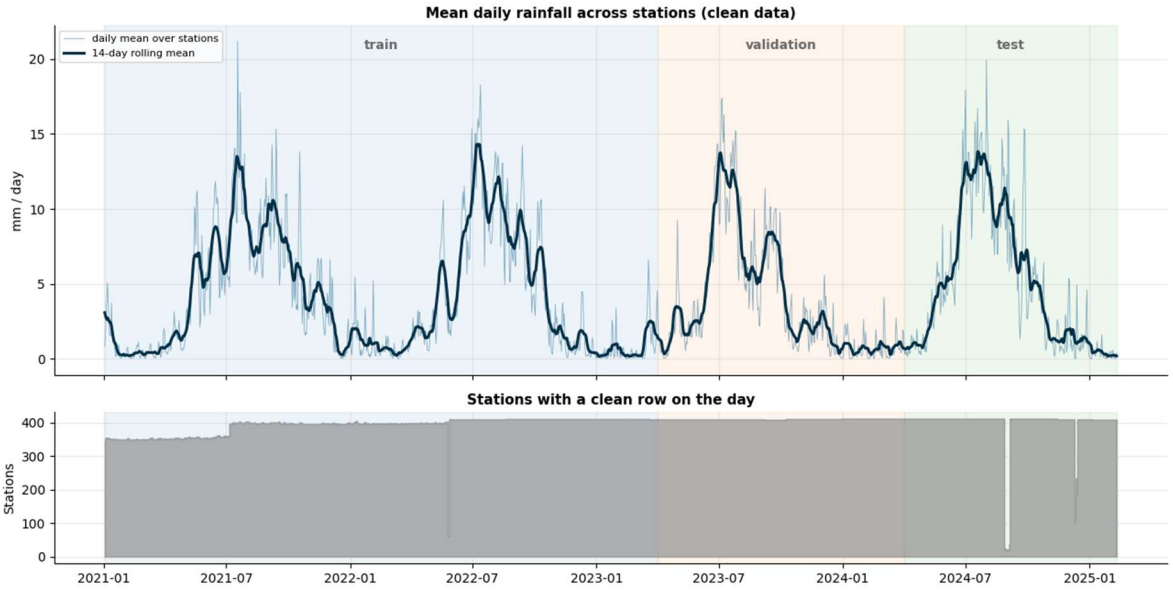


Figure 2. Daily rainfall series and the chronological training, validation, and test periods.

Interpretation: The timeline separates training, validation, and final testing in chronological order. This better reflects forecasting future rainfall and reduces the risk that later observations influence model fitting or model selection.

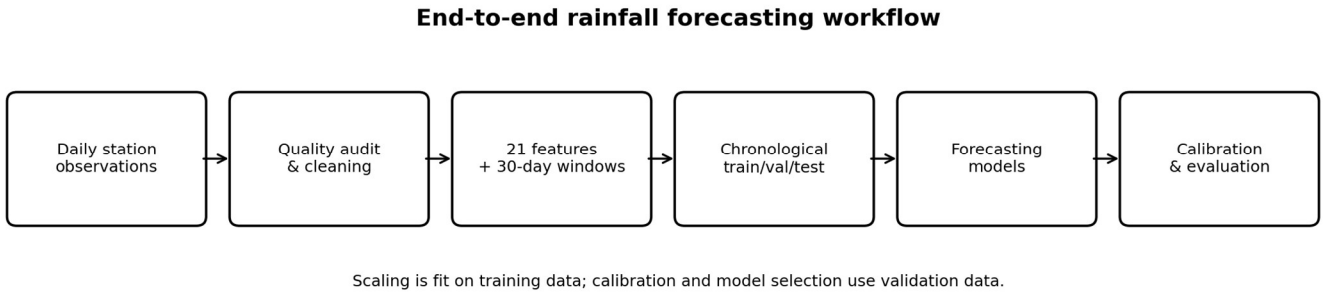


Figure 3. End-to-end workflow of the rainfall forecasting experiment.

Interpretation: The workflow makes the order of operations explicit: auditing and feature construction precede sequence modelling, while model selection and calibration rely on validation data. The held-out test set is reserved for the final performance report.

4. Feature Engineering and Sequence Construction

4.1 Transforming the Rainfall Target

Rainfall is non-negative, has many zero values, and includes rare large amounts. The model therefore learns the logarithmically transformed target rather than raw millimetres. For rainfall R in millimetres, the transformation is:

$$y_t = \log(1 + R_t) \quad \text{and} \quad \hat{R}_t = \max(\exp(\hat{y}_t) - 1, 0)$$

log1p compresses very large rainfall values; expm1 converts the model output back to millimetres.

The logarithmic transformation reduces the influence of extreme values during optimization and allows the model to learn from dry and light-rain days without the largest events dominating the loss. The final evaluation is nevertheless reported in millimetres so that the errors have a direct physical interpretation.

4.2 The 21 Input Features

Feature group	Features	Purpose
Weather and derived weather (6)	avg_temp, min_temp, max_temp, temp_range, wind_speed, air_pressure	Describe the local daily atmospheric state and temperature range.
Rain history (1)	rain_log = log1p(rainfall)	Represent recent precipitation while reducing skew.
Station geography (3)	latitude, longitude, elevation	Encode spatial position and elevation-related differences.
Neighbour / regional context (9)	dp_own, nb_lr, nb_lrmax, nb_dp, nb_dT, nb_wind, reg_lr, nat_lr, nat_dp	Represent local pressure tendency, nearby-station rainfall and weather,

		regional-cell averages, and national context.
Seasonality (2)	doy_sin, doy_cos	Represent the annual cycle without a discontinuity at the end of the year.

The 9 regional features are computed from the wider station network. The pipeline identifies the six nearest stations using geographic coordinates adjusted by latitude, then computes summary information such as mean and maximum log rainfall at neighbouring stations, neighbouring pressure/temperature tendencies, and regional/national averages. The geographic grid uses cells of approximately 3 degrees. Missing regional summaries are filled using medians calculated from training rows only.

4.3 Cyclical Calendar Encoding

Month or day-of-year can be represented cyclically so that the end and beginning of a year remain close in feature space. It uses the day of year d and a period of 365.25 days:

$$\text{doy_sin} = \sin(2\pi d / 365.25) \quad ; \quad \text{doy_cos} = \cos(2\pi d / 365.25)$$

Using sine and cosine avoids treating day 365 and day 1 as distant points. These features allow the networks to learn recurring seasonal patterns alongside recent weather observations.

4.4 Standardization Without Test Leakage

Continuous features are standardized using the mean and standard deviation computed on training data only. The fitted scaler is then reused unchanged for validation and test observations. This prevents information from the later periods from influencing the feature scaling.

$$z = (x - \mu_train) / \sigma_train$$

μ_train and σ_train are estimated only from the training portion.

4.5 Creating Sliding Windows

After sorting by station and date, the pipeline constructs one sample from 30 consecutive daily rows belonging to the same station. The target is the following day's log-transformed rainfall. A window is rejected if any day is missing from the consecutive sequence, if the station changes inside the window, or if the target is outside the correct chronological split.

$$X_i \in \mathbb{R}^{30 \times 21} \quad ; \quad y_i = \log(1 + \text{rainfall on target day})$$

Each sample contains 30 time steps and 21 input features.

Window set	Number of samples	Target date range	Mean target rainfall (mm)
Training	297,046	2021-02-01 – 2023-03-31	4.094
Validation	150,204	2023-04-01 – 2024-03-31	3.738
Test	101,540	2024-04-01 – 2025-02-10	5.049

The number of valid windows is lower than the number of cleaned rows because each sample requires a 30-day history and a next-day target. Removed days break a sequence, so windows around gaps cannot be used. The final evaluation contains 101,540 predictions from 412 stations.

5. Exploratory Data Analysis

5.1 Rainfall Distribution and Imbalance

The exploratory analysis confirms that the target distribution is zero-inflated and heavy-tailed. Dry days form a large portion of all station-days, while the amount of rainfall on wet days is concentrated in lower categories with a small number of very wet observations. This imbalance makes a single regression score insufficient: a model can perform reasonably on common dry/light-rain days but still underestimate rare heavy events.

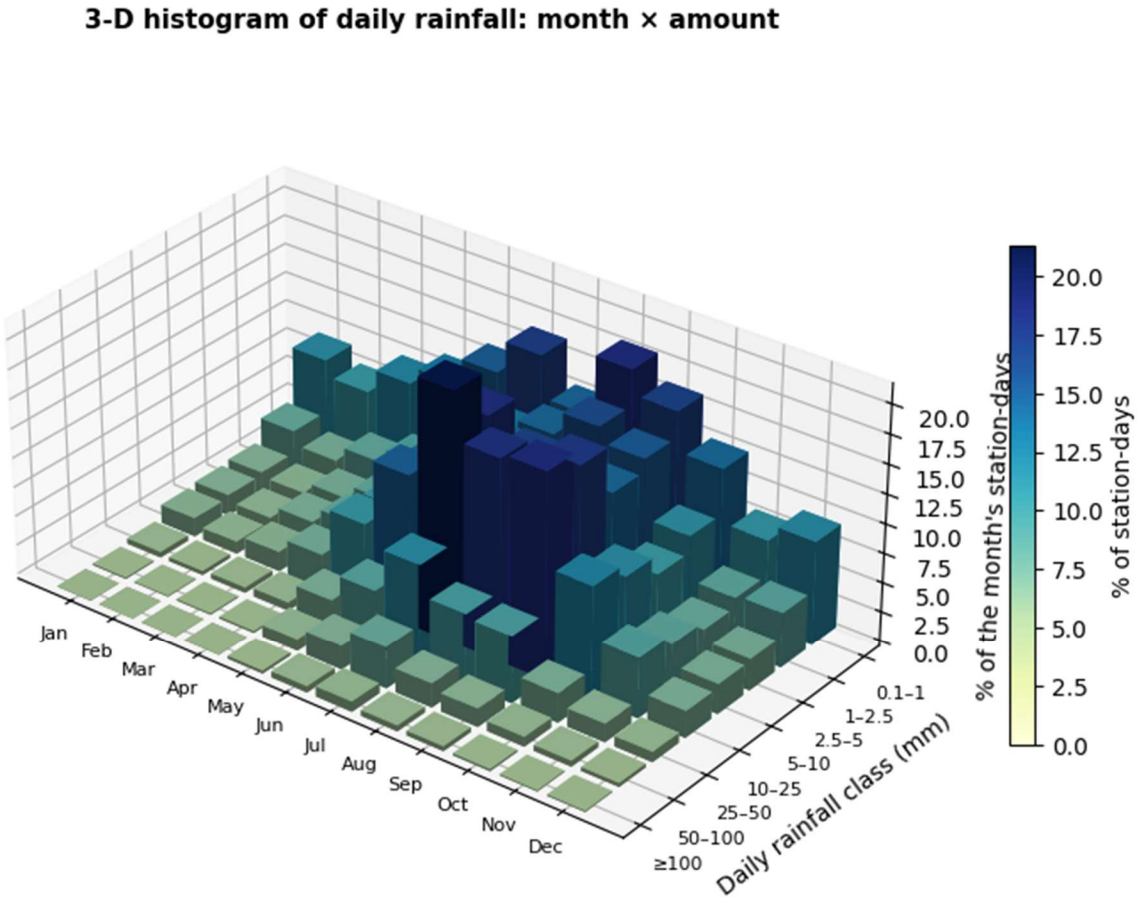


Figure 4. Rainfall amount distribution and seasonal composition in the cleaned modelling data.

Interpretation: Rainfall amounts are strongly right-skewed: many station-days have little or no rain, while a small number have very large totals. The log1p transformation reduces the influence of extreme values during training, while event metrics separately assess whether rain occurs.

5.2 Seasonality and Geography

Rainfall differs substantially between stations and seasons. Latitude, longitude, and elevation are included because station-level rainfall regimes vary across India. The annual cycle is encoded using day-

of-year sine and cosine features, while neighbouring and regional rainfall features represent whether nearby locations are also experiencing precipitation.

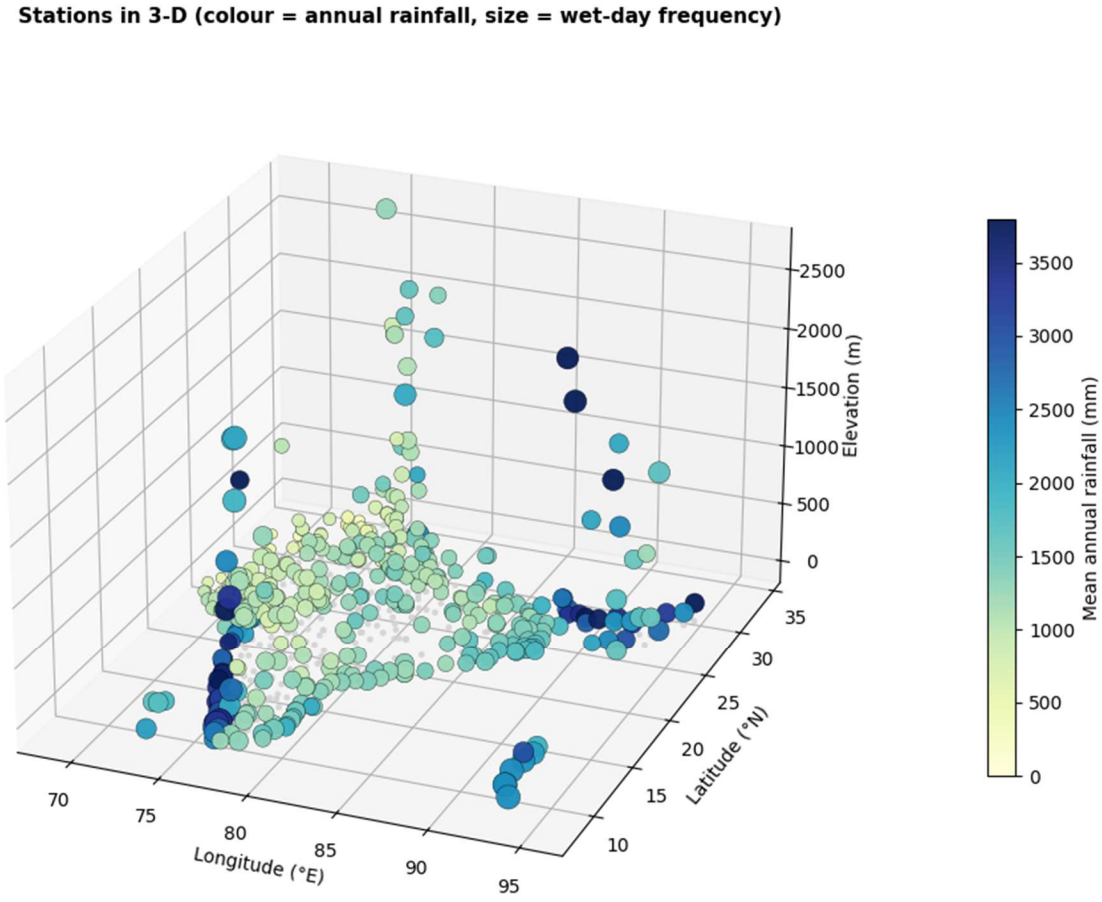
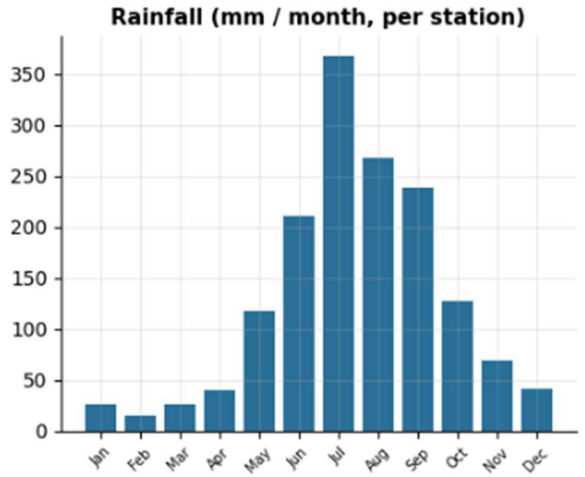
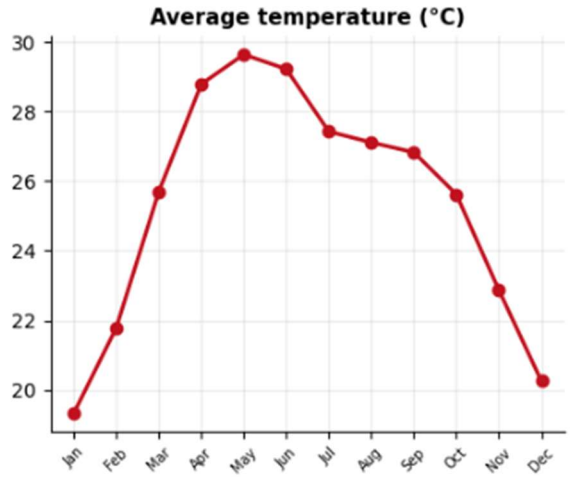


Figure 5. Spatial distribution of rainfall across stations.

Interpretation: Station-level rainfall differs across India, reflecting geographic and climatic diversity. This supports evaluating performance by station as well as with a pooled score, because a single aggregate metric can hide regional weaknesses.





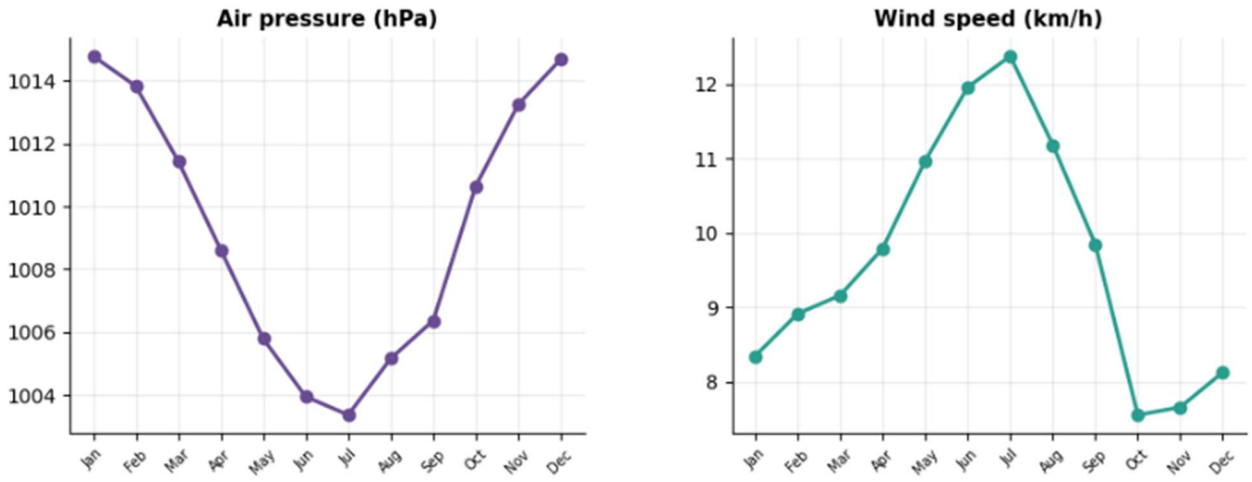
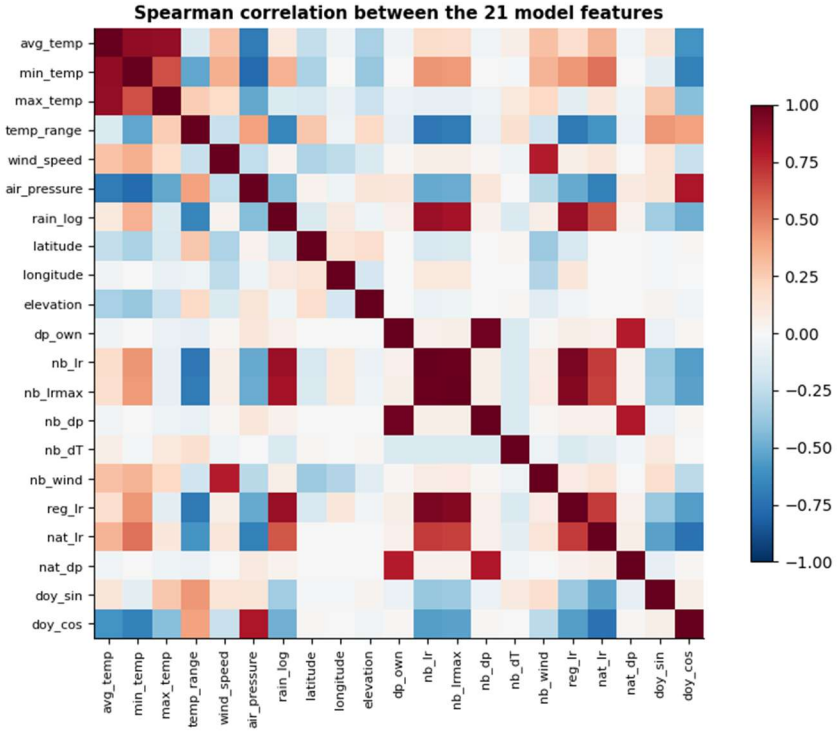


Figure 6. Seasonal and latitude-related rainfall patterns.

Interpretation: The seasonal patterns show that rainfall varies across the year and by location. Sine/cosine day-of-year features encode the annual cycle continuously, avoiding an artificial discontinuity between the end and beginning of the calendar year.

5.3 Feature Relationships

The analysis examines Spearman correlations between input features and the next-day log-rainfall target on training windows. Recent rainfall and neighbouring/regional rainfall features are among the strongest signals. Temperature range, pressure, and seasonal terms also contain information, while wind, elevation, and some day-to-day tendencies have weaker standalone relationships. Several features are correlated with one another, such as average/minimum/maximum temperature and overlapping neighbour/regional rainfall summaries. This supports evaluating compact sequence models with a consistent input set rather than assuming every feature contributes independently.



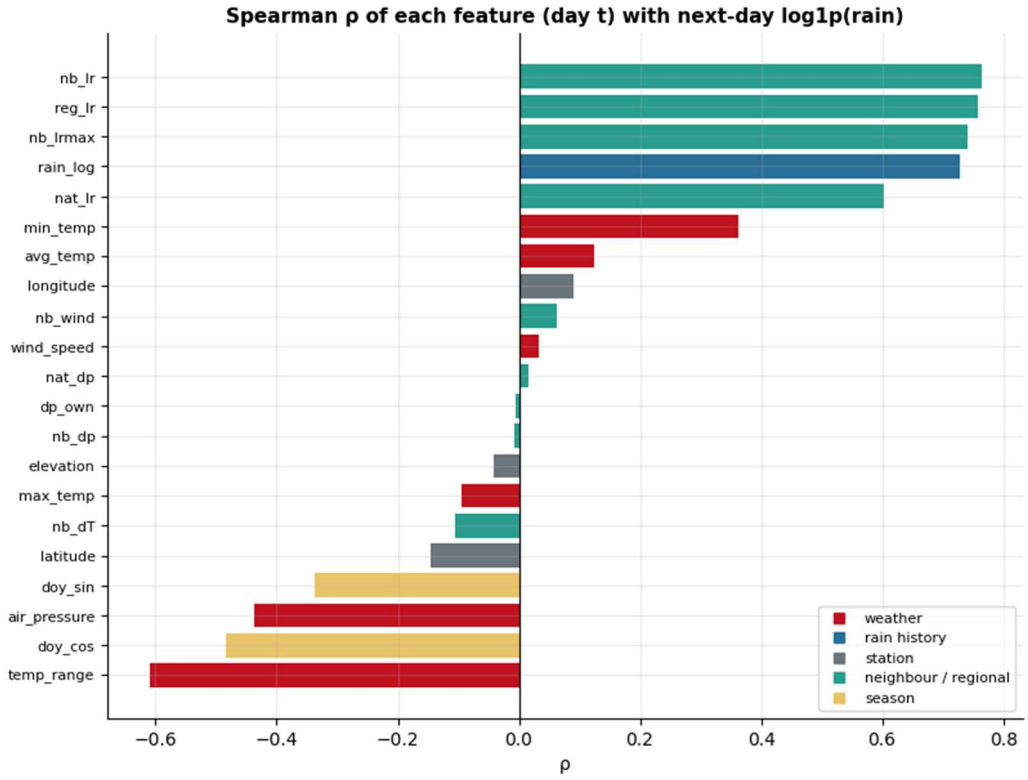


Figure 7. Correlation structure among the input features and next-day rainfall target.

Interpretation: Correlation patterns provide a first check of relationships among weather variables, engineered regional signals, and the target. Correlation is descriptive rather than causal, and weak pairwise correlation does not rule out useful nonlinear or time-dependent information.

Correlation is descriptive, not proof of causality. A feature can have a weak individual correlation yet still contribute through interactions with other variables or through its temporal sequence.

6. Forecasting Methodology

The forecasting pipeline has five main stages: (1) audit and clean daily station observations; (2) engineer weather, geographic, seasonal, and regional features; (3) standardize continuous features using training-only statistics and construct 30-day windows; (4) train the three neural models and the gradient-boosting reference using training windows, selecting checkpoints with validation loss; and (5) estimate calibration factors on validation predictions and report all final metrics on the untouched test targets.

Stage	What was done	Output
Data preparation	Filter, clean, sort, and identify stations; preserve chronological partitions.	Clean daily station table.
Feature engineering	Create 21 inputs including log rainfall, cyclical dates, and regional summaries.	Standardized feature matrix.
Sequence generation	Build 30-day same-station consecutive windows for next-day prediction.	Train/validation/test samples.
Model training	Train BiLSTM, BiLSTM-PINN, and Transformer; use AdamW, learning-rate scheduling, early stopping, and best-validation checkpoints.	Trained neural models.

Calibration and evaluation	Fit one multiplicative rainfall calibration factor per neural model on validation data; evaluate on test in mm.	Comparable model scores and plots.

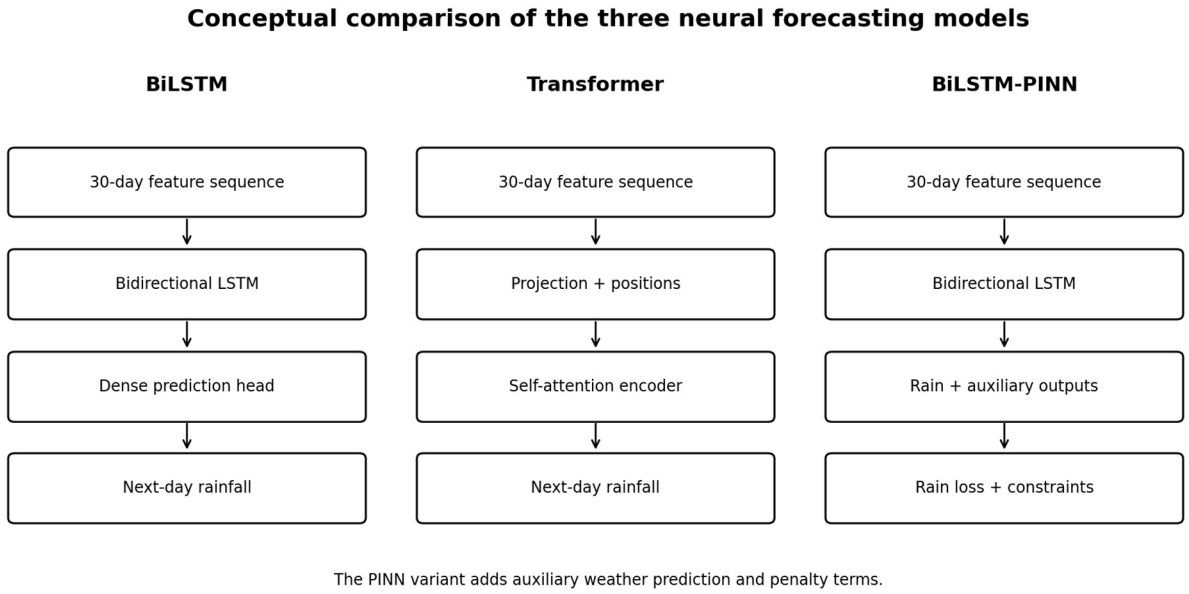


Figure 8. Conceptual architecture of BiLSTM, Transformer, and BiLSTM-PINN models.

Interpretation: BiLSTM and Transformer learn temporal representations through different mechanisms—recurrent processing versus self-attention. BiLSTM-PINN uses the recurrent backbone but adds auxiliary weather outputs and penalty terms to encourage physically plausible predictions.

6.1 BiLSTM Regressor

The BiLSTM processes the 30 time steps in forward and reverse directions within the historical input window. Its final forward and backward hidden states are concatenated and passed through a regression head consisting of layer normalization, a dense layer, GELU activation, dropout, and a final output unit. The model uses two recurrent layers, 64 hidden units per direction, dropout of 0.30, and a 64-unit prediction head in the supplied configuration.

$$h = [h_{forward} ; h_{backward}] \quad ; \quad \hat{y} = f\theta(X)$$

The output $\hat{y}$ estimates log1p of next-day rainfall.

Bidirectionality does not mean the model observes future calendar days. Both directions operate only over the known 30-day input window; the target day remains outside that window.

6.2 Transformer Regressor

The Transformer first projects each 21-dimensional feature vector into a 64-dimensional representation, adds sinusoidal positional encodings, and processes the sequence with a two-layer Transformer encoder. The supplied configuration uses four attention heads, a feed-forward dimension of 128, and dropout of 0.20. The model concatenates the encoded representation of the last day with the mean representation over the whole sequence and passes the result through a dense regression head.

$$\text{Attention}(Q,K,V) = \text{softmax}(QK^T / \sqrt{d_k})V$$

Self-attention assigns weights to relationships between positions in the 30-day sequence.

The attention mechanism can combine information from multiple days rather than relying only on recurrent state updates. However, the model's performance must be determined empirically; a more complex architecture is not guaranteed to outperform a recurrent model.

6.3 BiLSTM-PINN: Physics-Informed Auxiliary Learning

The BiLSTM-PINN uses the same recurrent backbone as the BiLSTM but produces six outputs: one log-rainfall prediction and five auxiliary next-day weather-state predictions (average temperature, minimum temperature, maximum temperature, wind speed, and air pressure). The auxiliary targets encourage the shared representation to learn a coherent next-day weather state. Additional penalty terms discourage outputs that violate selected physical expectations.

The total loss combines rainfall prediction error, auxiliary weather-state error, and weighted physics residuals:

$$L_{total} = L_{rain} + \lambda_{aux} L_{aux} + \lambda_{phys} L_{phys}$$

$$L_{rain} = \text{MSE}(\hat{y}_{log\text{-}rain}, y_{log\text{-}rain}) \quad ; \quad L_{aux} = \text{MSE}(\hat{s}_{next}, s_{next})$$

Three physics residual groups are used. First, rainfall in log1p space and wind speed should be non-negative. Second, the predicted temperature state should satisfy minimum temperature ≤ average temperature ≤ maximum temperature. Third, one-day changes in average temperature and pressure are penalized when their absolute magnitude exceeds plausible thresholds estimated from the 99.5th percentile of consecutive-day changes in the training data.

$$L_{phys} = w_{nonneg} R_{nonneg} + w_{order} R_{order} + w_{tend} R_{tend}$$

Sets $\lambda_{aux} = 0.3$, $\lambda_{phys} = 0.1$, and each residual weight $w = 1$.

For the temperature-order constraint, the penalty increases only when the predicted minimum exceeds the predicted average, or the predicted average exceeds the predicted maximum. For tendency constraints, only changes beyond the training-derived threshold are penalized. These are soft constraints: they influence training through the loss and do not guarantee that every prediction is physically perfect.

6.4 Reference Models

Two simple baselines are used. The climatology baseline predicts the mean rainfall for the station and calendar month using training observations; if a station-month mean is unavailable, it falls back to the overall training month mean and then the global training mean. The persistence baseline predicts tomorrow's rainfall using rainfall on the last observed input day.

A histogram gradient-boosting regressor is also trained on the same target windows as a tabular reference. Its features include the last four days of standardized inputs and summaries of rainfall history across the 30-day window. It uses up to 400 boosting iterations, a learning rate of 0.06, at most 31 leaf nodes, and a minimum of 100 samples per leaf. The best iteration is selected by validation loss.

7. Training Protocol and Calibration

7.1 Optimization and Regularization

Setting	Value / procedure
Random seed	42
Batch size	512 for training; 2,048 for validation and test loaders
Maximum epochs	40; training may stop earlier
Early stopping	Patience 6 epochs; minimum validation improvement 0.0001 in log-rain MSE
Optimizer	AdamW with weight decay 0.05
Learning rates	BiLSTM/BiLSTM-PINN: 5×10^{-4}; Transformer: 3×10^{-4}
Learning-rate schedule	One warm-up epoch followed by cosine decay to 1% of the initial learning rate
Input-noise regularization	Gaussian noise with standard deviation 0.05 on standardized inputs during training
Gradient control	Gradient norm clipping at 1.0
Checkpoint selection	Restore the model weights with the best validation rainfall MSE.

The code uses fixed random seeds for Python, NumPy, and PyTorch. Training loss is the mean squared error on log1p rainfall. Validation loss is used for early stopping and checkpoint selection. The test set is not used to choose the checkpoint or the calibration multiplier.

7.2 Back-Transformation Calibration

Optimizing squared error in log space and then applying expm1 can systematically underestimate the mean rainfall in millimetres. To reduce this bias, a single multiplicative factor is estimated separately for each neural model from validation predictions and observations. Let y_i be observed rainfall in millimetres and p_i the uncalibrated prediction:

$$\alpha = \Sigma_i(y_i\, p_i) \,/\, \Sigma_i(p_i^2) \quad ; \quad p_{i,\mathrm{cal}} = \alpha\, p_i$$

The fitted factors are approximately 1.63 for BiLSTM, 1.89 for BiLSTM-PINN, and 1.66 for Transformer. The same validation-derived factor is applied to test predictions. Calibration improves the rainfall amount estimates, but it does not alter the model ranking by itself in a way that should be assumed in advance; all results are measured after applying the stated procedure.

8. Evaluation Metrics

Because the task combines rainfall amount estimation with rain-event identification, the report uses regression metrics in millimetres and classification-style metrics for a rainy day. For rain-event metrics, a day is labelled rainy when observed rainfall is at least 1 mm. The prediction threshold for binary decisions is selected using validation data only.

8.1 Regression Metrics

$$\mathrm{RMSE} = \sqrt{[(1/n)\, \Sigma_i\, (p_i - y_i)^2]}$$

Root Mean Squared Error; penalizes large errors more strongly.

$$\mathrm{MAE} = (1/n)\, \Sigma_i\, |p_i - y_i|$$

Mean Absolute Error; average absolute error in millimetres.

$$R^2 = 1 - [\Sigma_i(p_i - y_i)^2 / \Sigma_i(y_i - \bar{y})^2]$$

Coefficient of determination; values closer to 1 indicate better fit relative to the observed mean.

$$\text{Bias} = (1/n)\ \Sigma_i(p_i - y_i)$$

Positive bias indicates overprediction; negative bias indicates underprediction.

Pearson correlation (PCC) measures linear association between observed and predicted amounts. Spearman correlation (SCC) measures rank association and is useful when the exact amount is difficult to estimate but the relative ordering is informative. Neither correlation alone measures the absolute error, so both are reported with RMSE and MAE.

8.2 Rain-Event Metrics

$$\text{Precision} = TP / (TP + FP) \quad ; \quad \text{Recall} = TP / (TP + FN)$$

$$F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

Here, TP is a correctly predicted rainy day, FP is a predicted rainy day that was actually dry, and FN is a rainy day missed by the model. ROC-AUC measures how well the continuous predictions rank rainy days above dry days across possible thresholds. AUC and F1 complement the amount-based metrics, but they do not replace them.

9. Results

9.1 Main Test-Set Comparison

The final test set contains 101,540 predictions covering 412 stations from 1 April 2024 to 10 February 2025. The table below reports the calibrated neural predictions alongside the two baselines and the gradient-boosting reference. Lower RMSE and MAE are better; higher R^2, PCC, SCC, AUC, and F1 are generally better. Bias closer to zero indicates less systematic over- or underprediction.

Model	RMSE mm	MAE mm	R^2	PCC	SCC	AUC rain	F1 rain	Bias mm
Climatology	11.588	5.224	0.230	0.480	0.658	0.856	0.731	+0.058
Persistence	12.871	4.759	0.050	0.522	0.722	0.868	0.773	−0.020
Gradient Boosting	11.032	3.777	0.302	0.609	0.786	0.927	0.809	−2.271
BiLSTM (calibrated)	10.521	4.072	0.365	0.606	0.784	0.926	0.809	−0.280
BiLSTM-PINN (calibrated)	10.491	4.070	0.369	0.609	0.783	0.926	0.809	−0.266
Transformer (calibrated)	10.529	3.940	0.364	0.606	0.785	0.927	0.811	−0.751

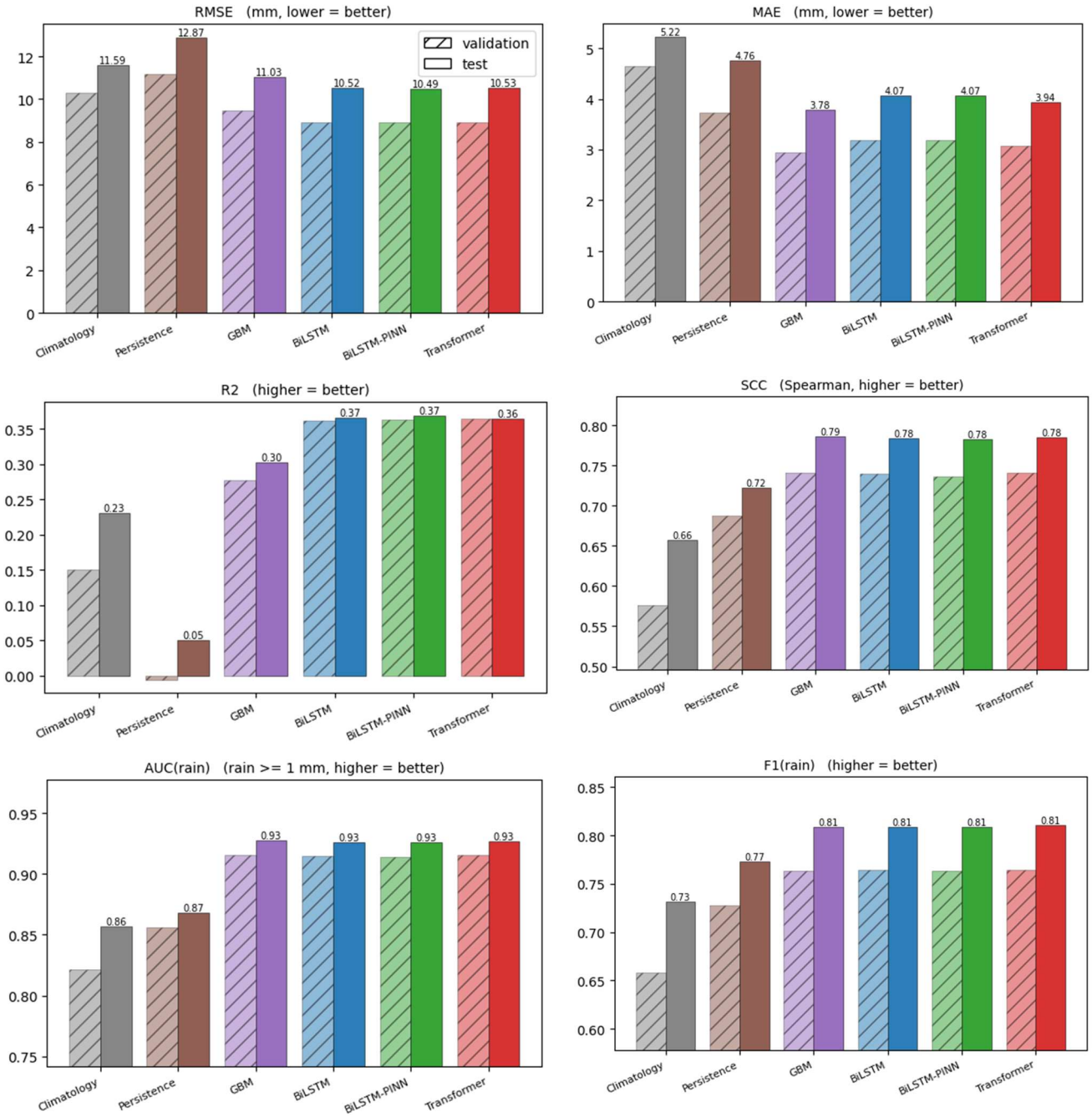


Figure 9. Validation and test comparison across regression and rain-event metrics.

Interpretation: The neural models improve test RMSE and R^2 over the simple reference methods, while differences among the neural variants are small. BiLSTM-PINN has the lowest reported RMSE, but the bootstrap interval for its difference from plain BiLSTM includes zero, so the improvement is not conclusive.

Among the calibrated neural models, BiLSTM-PINN has the lowest RMSE (10.491 mm) and the highest R^2 (0.369), but its advantage over calibrated BiLSTM (10.521 mm) and Transformer (10.529 mm) is small. Transformer has the highest reported rain-event F1 (0.811) and AUC (0.927). The gradient-boosting reference has a lower MAE (3.777 mm) than the calibrated neural models, despite its higher

RMSE. This indicates that the error distributions differ: RMSE is more sensitive to a small number of large misses.

The test RMSE of calibrated BiLSTM-PINN is about 9.5% lower than climatology: (11.588 − 10.491) / 11.588 × 100 ≈ 9.5%. It is about 18.5% lower than persistence: (12.871 − 10.491) / 12.871 × 100 ≈ 18.5%. These comparisons describe this test period and dataset processing setup; they do not establish universal superiority for all regions or future periods.

9.2 Calibration Effect

Before calibration, the neural models have test RMSE values of 11.051 mm for BiLSTM, 11.334 mm for BiLSTM-PINN, and 11.203 mm for Transformer. After applying the multipliers fitted on validation data, the corresponding RMSE values become 10.521, 10.491, and 10.529 mm. Calibration also reduces the magnitude of the mean negative bias, especially for BiLSTM and BiLSTM-PINN. It is therefore an important component of the final prediction pipeline, not merely a reporting adjustment.

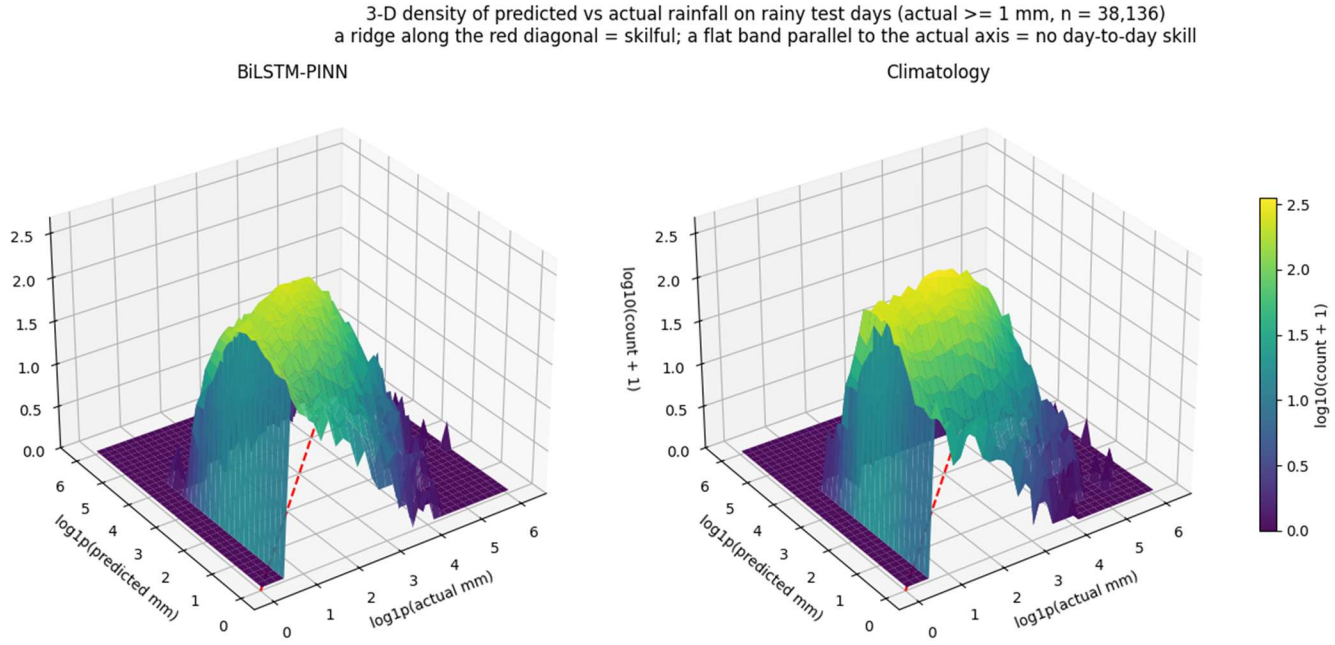


Figure 10. Predicted versus observed rainfall distributions for the neural models.

Interpretation: Comparing predicted and observed distributions reveals whether forecasts reproduce the overall scale and spread of rainfall, not just average error. Validation-based calibration adjusts the prediction scale, but cannot fully eliminate errors on rare extreme-rainfall days.

9.3 Rainy-Day Classification

Rain/no-rain evaluation defines an event as rainfall of at least 1 mm. On the test set, the calibrated models reach AUC values of 0.926–0.927 and F1 values of 0.809–0.811. These scores indicate strong ranking performance for separating rainy and dry days relative to the baseline rate. The Transformer obtains the highest F1, while BiLSTM and BiLSTM-PINN are very close. Because the binary threshold is selected on validation data, the test scores are not obtained by tuning a threshold directly on the test set.

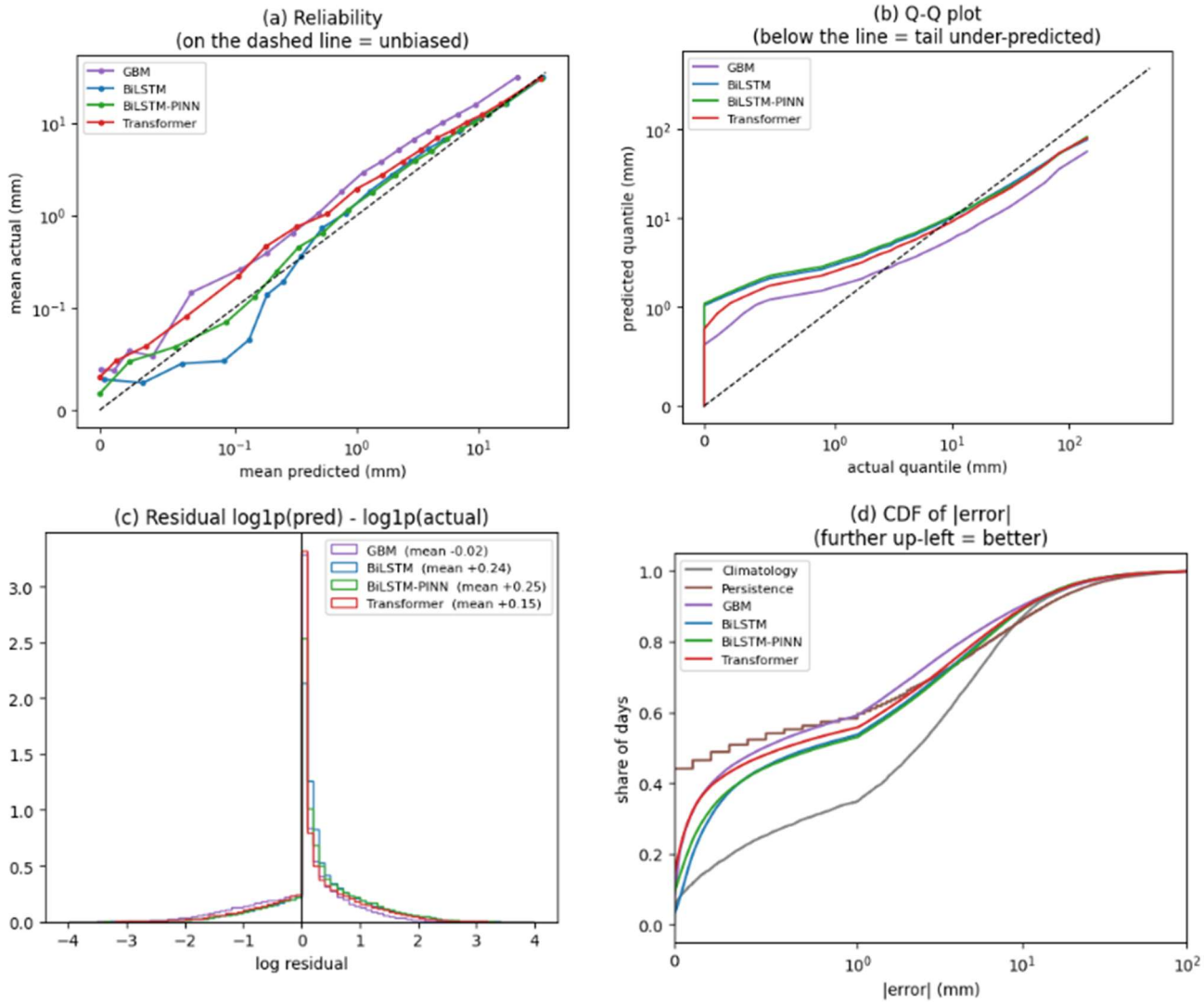


Figure 11. Rain-event reliability, ROC/precision-recall and classification analysis.

Interpretation: These plots assess rainfall occurrence as a classification task using the 1 mm event threshold. ROC/AUC and precision-recall/F1 capture different aspects of event detection, so they complement amount-based RMSE and MAE.

9.4 Error by Rainfall Intensity

The test observations are grouped into six intensity classes: dry (<1 mm), light (1–5 mm), moderate (5–20 mm), heavy (20–50 mm), very heavy (50–100 mm), and extreme (>100 mm). This breakdown is important because a model can score well overall by fitting common dry and light-rain days while underestimating the upper tail. The reports that only 299 test station-days are in the extreme category above 100 mm. Performance estimates for such rare events are therefore less stable and should be interpreted cautiously.

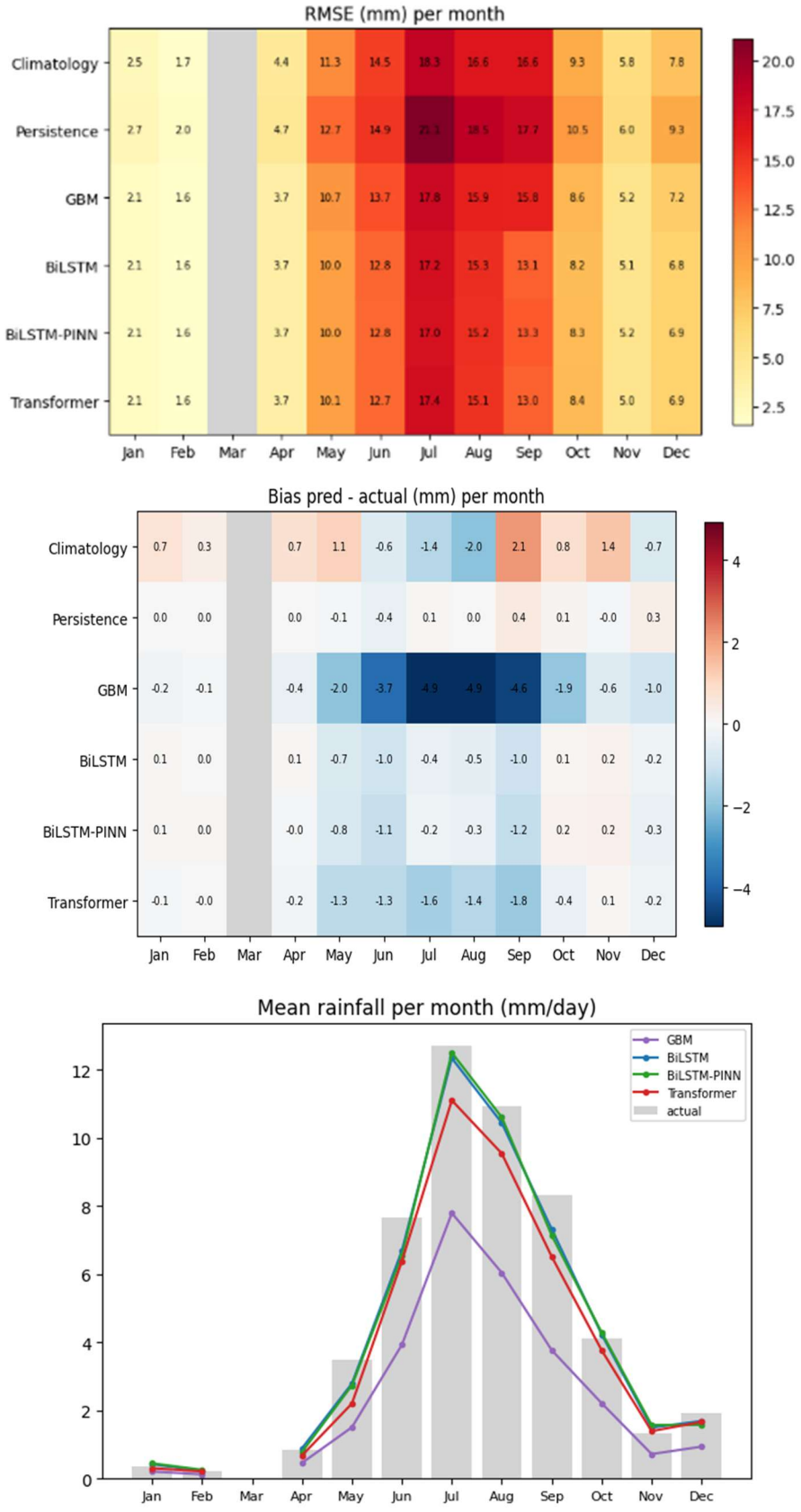


Figure 12. Error and bias patterns across months on the held-out test set.

Interpretation: Monthly error and bias reveal that model performance changes across the annual cycle. This variation can indicate seasonal distribution shifts and helps identify periods when a single pooled test metric may overstate reliability.

9.5 Station-Level Performance

Station-wise evaluation is calculated for stations with at least 30 test observations. The best neural model by overall test RMSE is calibrated BiLSTM-PINN. It has lower station-level RMSE than climatology at 95% of eligible stations and lower station-level RMSE than persistence at 95% of eligible stations. Across these stations, the median station-level correlation is 0.59, median RMSE is 7.92 mm, and median absolute bias is 0.55 mm.

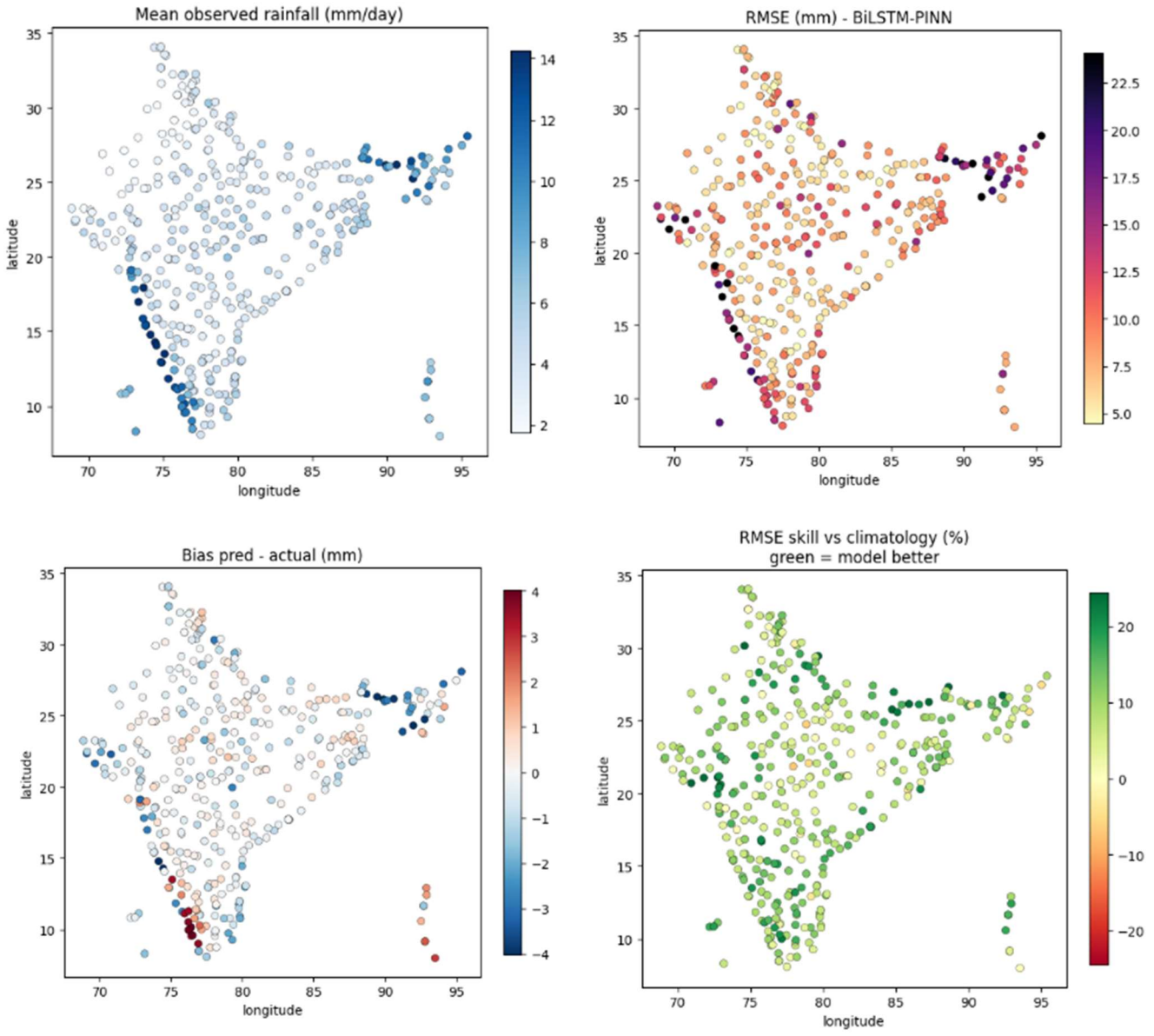


Figure 13. Geographic variation in station-level rainfall error and bias.

Interpretation: Station-level diagnostics show how forecast errors vary geographically rather than assuming all stations perform alike. Median station statistics are useful summaries, but should be read alongside the spread and the number of observations available at each station.

The national summary therefore conceals substantial local differences. A forecast system intended for operational use should show station-level performance and uncertainty rather than relying only on one aggregate score.

9.6 Additional Spatial and 3-D Diagnostics

The analysis also generates 3-D views to examine the joint relationships among rainfall, location, and model performance. These visualizations are exploratory diagnostics: they help identify geographic clusters, seasonal differences, and where forecast errors are larger, but they are not additional independent validation experiments.

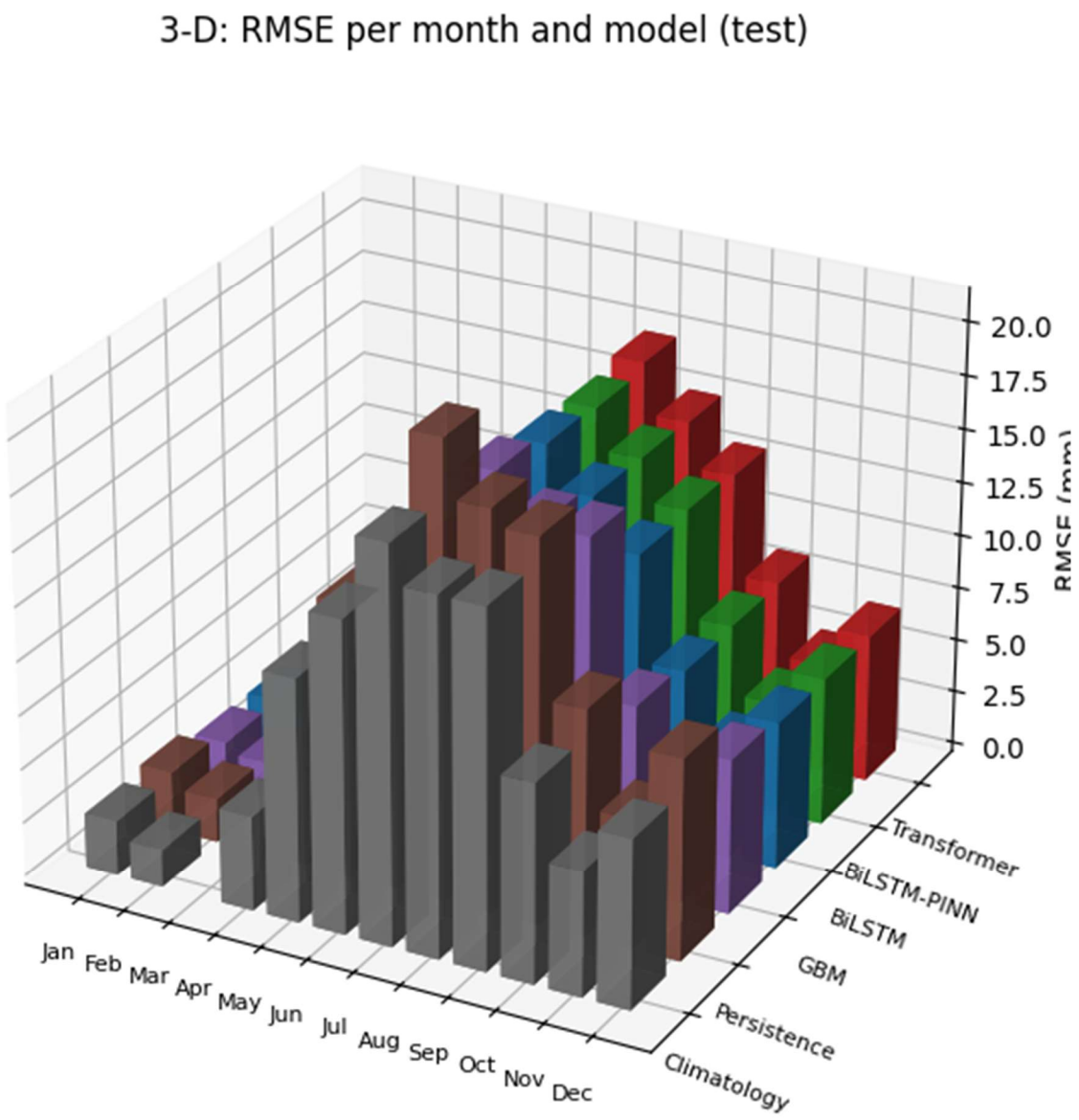


Figure 14. Three-dimensional diagnostics comparing rainfall predictions and station/month-level performance.

Interpretation: The 3-D views are exploratory diagnostics for comparing observations, predictions, and spatial or temporal patterns. They can reveal broad structure, but exact model ranking should rely on held-out numerical metrics rather than visual impression alone.

10. Discussion

10.1 What the Experiments Show

- Sequence models outperform the climatology and persistence baselines on test RMSE after the stated validation calibration procedure.
- BiLSTM-PINN has the best calibrated neural-model RMSE, but the 0.030 mm difference from BiLSTM is small and should not be described as a decisive architecture win.
- Transformer performs best on the reported F1 rain-event score (0.811), though its margin over the other neural models is only 0.002.
- Gradient boosting is competitive and achieves the lowest MAE in the reported test comparison. Neural models do not dominate every metric.
- Rainfall intensity and station location matter: the rarest, highest-rainfall events remain substantially harder than common conditions.

10.2 Interpreting the BiLSTM-PINN Result

Physics-informed auxiliary learning provides a principled way to regularize the model with simple weather-state constraints. In this experiment, however, the gain is modest: calibrated BiLSTM-PINN reduces RMSE by only 0.030 mm compared with calibrated BiLSTM. The block-bootstrap comparison gives a 95% interval for the RMSE difference (BiLSTM-PINN minus BiLSTM) of approximately −0.089 to +0.027 mm, which includes zero. Therefore, the observed RMSE difference is not conclusive evidence that the PINN variant is statistically superior.

The bootstrap procedure resamples test calendar days in blocks/units and computes uncertainty intervals for RMSE and MAE. It is a useful uncertainty check because multiple station predictions on the same date may share weather conditions. The report consequently presents BiLSTM-PINN as the best point estimate among neural models, while acknowledging that the difference is small and uncertain.

10.3 Why RMSE and MAE Tell Different Stories

RMSE squares each error before averaging and therefore gives extra weight to large misses. MAE weights each absolute error linearly. Gradient boosting's lower MAE but higher RMSE than the calibrated neural models suggests that it makes smaller errors on many cases but has some larger errors that affect RMSE. This is a key reason to report several complementary metrics instead of choosing a model from one score alone.

11. Limitations and Threats to Validity

- Missing-data selection: excluding the 2015–2020 period avoids a clear missingness bias, but it also reduces the historical period available for training. Results apply to the selected data from 2021 onward.
- Incomplete spatial coverage: the original resource description reports 458 stations, whereas cleaning leaves 414 station identifiers and the test set covers 412. The results should not be generalized to stations absent from the retained data.
- Test-period composition: the test set ends on 10 February 2025 and contains no March observations. Seasonal comparisons must account for the unequal month coverage.
- Rare extremes: only 299 test observations exceed 100 mm, so aggregate scores cannot fully characterize extreme-event skill.

- Small differences between neural models: bootstrap intervals overlap, and the BiLSTM-PINN RMSE difference relative to BiLSTM is not conclusive.
- Regional features and deployment availability: the model uses neighbouring and regional observations. A real-time implementation must ensure those observations are available at forecast time and that feature generation does not use information from the target day.
- Calibration dependence: the reported calibrated scores rely on a scalar estimated from the validation period. Performance may change under a different climate regime or a later operational period.
- No probabilistic uncertainty output: the current implementation primarily produces point forecasts. Users do not receive prediction intervals from these model outputs.

12. Conclusion and Future Work

This research implements a complete daily rainfall forecasting workflow, from data auditing and cleaning to feature engineering, time-series window generation, neural model training, calibration, and test evaluation. The study uses 597,294 cleaned station-day rows to build 297,046 training windows, 150,204 validation windows, and 101,540 test windows. Every model predicts rainfall one day ahead from the preceding 30 consecutive days at a station, using 21 inputs that include weather measurements, station geography, seasonal encoding, and regional context.

On the held-out test period, calibrated BiLSTM-PINN obtains the lowest RMSE among the neural models at 10.491 mm and R^2 of 0.369. Calibrated BiLSTM and Transformer are close, while Transformer achieves the highest rain-event F1 at 0.811. All calibrated neural models outperform climatology and persistence in RMSE, but gradient boosting remains competitive and has the lowest MAE. The small difference between BiLSTM-PINN and BiLSTM, together with the bootstrap interval, means that the results support a modest point-estimate advantage rather than a definitive claim of statistical superiority.

Future work should focus on the following directions:

- Use rolling-origin or multi-period temporal evaluation to test stability across different years and seasonal regimes.
- Improve heavy and extreme rainfall prediction through intensity-aware objectives, probabilistic outputs, and event-based evaluation.
- Investigate feature ablation to quantify the contribution of neighbouring-station and national/regional variables.
- Check the robustness of the PINN constraints and their weights, and compare them against the same BiLSTM backbone over multiple seeds.
- Report station-specific uncertainty and operational measures such as latency, data availability, and forecast update reliability.
- Validate all regional features in a real forecasting setup to ensure that only information available by the forecast issue time is used.

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5. Project implementation and experimental outputs: `DataAnalysi  3D.ipynb` and `rainfall-bilstm-transformer-1 (3).ipynb`, supplied with this project.